

Abstract

Acknowledgements

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Chapter 1

Introduction

Chapter 2

Theoretical Foundations

Chapter 3

Literature Review

3-1 LLMs and Financial NLP

3-1-1 A Survey of Large Language Models for Financial Applications: Progress, Prospects and Challenges [1]

Research Focus

The survey reviews how LLMs are applied in finance, emphasizing their role in:

- Sentiment analysis
- Time-series forecasting
- Financial reasoning and decision support
- Agent-based modeling for trading and simulation

It highlights both the potential and the challenges of adapting LLMs to financial NLP tasks.

Methodology

The authors conduct a structured literature review and classify applications into five major categories (see Table (3-1)). They also summarize widely used datasets and benchmarks.

Results

- LLMs outperform traditional models on financial NLP tasks.
- Domain-specific pretraining (e.g., FinBERT, FinGPT) further boosts accuracy.
- Benchmarks like FPB, FiQA, and FinQA confirm improvements in sentiment, QA, and reasoning.

Table (3-1) Application areas of LLMs in finance (adapted from [1])

Category	Examples
Linguistic tasks	Summarization of earnings calls, information extraction from filings
Sentiment analysis	Classifying financial news, reports, and microblogs
Financial time-series	Text-driven forecasting, anomaly detection
Financial reasoning	Q&A, decision support, numerical reasoning
Agent-based modeling	LLM-powered trading agents, market simulations

Limitations

- Data scarcity and domain adaptation difficulties
- Risk of look-ahead bias in financial modeling
- High computational cost and latency
- Lack of interpretability and explainability
- Legal, compliance, and regulatory risks

Contribution to My Project

3-1-2 FinBERT: Financial Sentiment Analysis with Pre-trained Language Models [2]

Research Focus

FinBERT addresses the challenge of sentiment analysis in financial texts, where general sentiment models often fail due to specialized jargon and domain-specific context. The goal is to adapt BERT for accurate classification of financial news, reports, and statements.

Methodology

- Pre-trains BERT on a large financial corpus (e.g., news, company filings, analyst reports).
- Fine-tunes on benchmark financial sentiment datasets for three-class classification (positive, negative, neutral).
- Compares performance with traditional ML and generic BERT sentiment models.

Table (3-2) FinBERT methodology overview

Step	Description
Pre-training	Domain-adaptive training on large financial text corpora
Fine-tuning	Sentiment classification (pos/neg/neu) using Financial PhraseBank and similar datasets
Baseline comparison	Benchmarked against lexicon-based and traditional ML approaches

Results

- Achieves significantly higher accuracy and F1-scores than generic sentiment models.
- Strong at capturing subtle financial expressions (e.g., “slight decline,” “robust outlook”).
- Demonstrates robustness across multiple datasets (e.g., Financial PhraseBank, FiQA).

Limitations

- Requires access to large high-quality financial text for pretraining.
- Still limited by BERT’s architecture (e.g., input length constraints).
- Like other transformers, lacks interpretability and explainability.
- Performance may degrade in contexts needing deep economic reasoning beyond textual sentiment.

Contribution to My Project

3-1-3 FinGPT: Open-Source Financial Large Language Models [3]

Research Focus

FinGPT aims to democratize access to financial large language models by providing an open-source, cost-effective, and customizable alternative to proprietary financial AI systems. It focuses on making LLMs accessible for tasks such as financial sentiment analysis, report summarization, and question answering.

Methodology

- Builds on open-source LLM backbones (e.g., GPT-style models).
- Applies financial domain adaptation through instruction tuning and lightweight fine-tuning.

- Uses reinforcement learning with human feedback (RLHF) to improve task alignment.
- Emphasizes efficiency by using low-rank adaptation (LoRA) and cost-effective training pipelines.

Table (3-3) FinGPT methodology highlights

Component	Description
Model backbone	Open-source GPT-family LLMs
Domain adaptation	Instruction tuning + financial data fine-tuning
Efficiency methods	LoRA, lightweight parameter-efficient training
Alignment	RLHF to ensure reliable outputs for financial tasks

Results

- Competitive performance on financial NLP benchmarks compared to proprietary models.
- Strong adaptability across tasks (sentiment analysis, QA, summarization).
- Open-source design enables transparency, customization, and collaboration.
- More cost-efficient than closed-source financial LLMs.

Limitations

- Dependent on the quality and coverage of financial training data.
- May underperform against large proprietary LLMs trained on massive datasets.
- Faces robustness challenges in high-stakes or compliance-heavy applications.
- Ethical, legal, and regulatory issues (e.g., risk of hallucinations in finance).

Contribution to My Project

3-2 RL Libraries and Reviews in Finance

3-2-1 Deep Reinforcement Learning in Quantitative Algorithmic Trading: A Review [4]

Research Focus

The paper explores the use of deep reinforcement learning (DRL) for low-frequency quantitative algorithmic trading. It frames the stock market as a complex, imperfect-information environment—similar to games like Go or Chess—where agents learn to maximize returns and control risk.

Methodology

- Focus restricted to automated, end-to-end DRL trading agents (minute-level to daily frequency).
- Excludes forecasting models, hybrid/metaheuristic approaches, portfolio optimization, and high-frequency trading.
- Categorizes DRL methods into three main families (see Table (3-4)).

Table (3-4) Categorization of DRL approaches in trading (adapted from [4])

Category	Examples
Critic-only (value-based)	Deep Q-Networks (DQN), Double DQN
Actor-only (policy-based)	REINFORCE, Policy Gradient methods
Actor-Critic hybrids	Advantage Actor-Critic (A2C), Deep Deterministic Policy Gradient (DDPG), PPO

Results

- DRL agents often outperform traditional baselines in simulations.
- Gains are typically *statistical*, not necessarily translating into real-world profitability.
- Current research remains proof-of-concept with modest results.

Limitations

- Most studies use simplified backtesting; lack real-time, live trading experiments.
- Very few comparisons across DRL variants or with human traders.
- Practical challenges: latency, transaction costs, and robustness under market frictions.
- Profitability in real-world conditions remains unproven.

Contribution to My Project

3-2-2 FinRL: A Deep Reinforcement Learning Library for Automated Stock Trading in Quantitative Finance [5, 6]

Research Focus

FinRL introduces one of the first open-source DRL libraries tailored to quantitative finance. Its main goal is to provide a complete, reproducible, and user-friendly framework that lowers the entry barrier for researchers and practitioners in automated trading.

Methodology

- Implements a **three-layer modular architecture**:
 - **Environment Layer**: Market simulation using historical/real-time data; integrates trading costs, liquidity, and risk aversion.
 - **Agent Layer**: State-of-the-art DRL algorithms (DQN, DDPG, PPO, SAC, A2C, TD3) with customizable reward functions.
 - **Application Layer**: Ready-to-run modules for stock trading, portfolio allocation, and cryptocurrency trading with backtesting tools.
- Built on PyTorch and OpenAI Gym-style APIs for accessibility and extensibility.
- Offers tutorials and step-by-step guides for education and reproducibility.

Table (3-5) Highlights of FinRL framework

Component	Key Features
Environment Layer	Historical/real-time data, transaction costs, liquidity constraints
Agent Layer	DRL algorithms (DQN, PPO, SAC, A2C, TD3), customizable rewards
Application Layer	Trading modules, portfolio strategies, crypto trading, backtesting

Results

- Demonstrates successful application of DRL agents across multiple trading tasks.
- Provides reproducible benchmarks and environments, enabling fair comparisons between algorithms.
- Widely adopted in research and education, bridging the gap between ML and finance.

Limitations

- Does not provide out-of-the-box profitable strategies — it is an infrastructure toolkit, not a trading solution.
- Some environment design choices (e.g., stock ordering in action space) may introduce subtle biases.
- Real-world deployment still requires significant domain expertise and strategy development.

Contribution to My Project

3-3 DRL for Financial Decision-Making

3-3-1 Deep reinforcement learning for automated stock trading: An ensemble strategy [7]

Research Focus

The paper proposes an ensemble deep reinforcement learning (DRL) strategy for automated stock trading. The aim is to combine the strengths of multiple DRL algorithms to create a more robust and adaptive trading agent capable of handling dynamic market regimes.

Methodology

- **Algorithms used:** PPO, A2C, and DDPG (all actor-critic based).
- **Load-on-demand technique:** Efficient data handling to reduce memory overhead in continuous action spaces.
- **Ensemble mechanism:**
 - Agents are trained on rolling windows of historical data.
 - A validation window selects the best agent based on Sharpe ratio and turbulence index.
 - The selected agent executes trades in the next trading period.
- **Dataset:** 30 Dow Jones Industrial Average (DJIA) stocks.
- **Baselines:** Compared against DJIA index and minimum-variance portfolio.

Table (3-6) Overview of ensemble strategy (Li et al., 2020)

Component	Description
Algorithms	PPO, A2C, DDPG (actor-critic DRL)
Selection metric	Sharpe ratio (risk-adjusted), turbulence index (risk control)
Process	Train → Validate → Select best agent → Trade
Benchmarks	DJIA index, min-variance portfolio

Results

- The ensemble strategy consistently outperforms single-agent DRL models (PPO, A2C, DDPG alone).
- Achieves higher Sharpe ratios and more stable risk-adjusted returns.
- Demonstrates adaptability by selecting agents that perform best in different market regimes.

Limitations

- Computationally expensive due to multiple agents and retraining.
- Performance depends heavily on window size, hyperparameters, and market conditions.
- Limited to three DRL algorithms — future work could expand to newer methods (e.g., SAC, TD3).
- Still evaluated only in backtesting; real-world trading deployment remains untested.

Contribution to My Project

3-3-2 A novel deep reinforcement learning based automated stock trading system using cascaded LSTM networks [8]

Research Focus

The paper addresses the difficulty of applying DRL to financial time series, which suffer from low signal-to-noise ratios and irregular patterns. It proposes a cascaded LSTM architecture to better capture temporal dependencies and latent structures in daily stock data.

Methodology

- **Two-stage cascaded LSTM:**
 - **Feature extraction stage:** First LSTM learns temporal patterns from raw daily stock data.
 - **Agent stage:** Extracted features are fed into a DRL agent (PPO) enhanced with another LSTM in its actor-critic networks.
- **Datasets:** Evaluated on four major indices — DJI (US), SSE50 (China), SENSEX (India), FTSE100 (UK).
- **Benchmarks:** Compared against buy-and-hold, PPO-MLP, standalone LSTM/MLP, LightGBM, gradient boosting, and ensemble DRL strategies.

Results

- Achieved performance improvements of **5–52%** across cumulative returns, max earning rate, and average profit per trade.
- Outperformed both traditional ML baselines and DRL models without cascaded LSTM.
- Showed particularly strong results in the SSE50 market, highlighting adaptability in emerging markets.

Limitations

- Increased complexity and training cost due to two-stage architecture.
- Sensitivity to hyperparameters, market regimes, and data frequency.
- Results based on backtesting only — no live trading validation.

Contribution to My Project

3-3-3 Commission Fee is not Enough: A Hierarchical Reinforced Framework for Portfolio Management [9]

Research Focus

This work highlights a key limitation in many DRL-based portfolio management systems: most models only consider commission fees as trading costs, neglecting the impact of **price slippage**, which can significantly alter execution outcomes. To address this, the authors propose **HRPM**, a hierarchical framework that separates portfolio allocation from trade execution.

Methodology

- **Two-tier hierarchical design:**
 - **High-level policy:** Allocates portfolio weights at lower frequency, aiming to maximize long-term returns.
 - **Low-level policy:** Executes trades within short windows, minimizing costs including slippage.
- **Training strategy:**
 - Pre-training both levels to establish baseline behaviors.
 - Iterative joint training for coordinated strategy–execution efficiency.
- **Evaluation:** Conducted on both U.S. and Chinese stock markets against DRL and traditional baselines.

Results

- **Trading costs:** Commission fees are only a minor part of total costs — slippage dominates, especially in large trades.
- **Entropy bonus:** Adding entropy to rewards diversifies portfolios.
 - $\eta = 0.05 \rightarrow$ highest annualized return (ARR).
 - $\eta = 0.1 \rightarrow$ lowest maximum drawdown (MDD).
- **Execution behavior:** The low-level policy spreads trades strategically — selling partially early and timing buys/sells with local price trends to reduce slippage.

- **Overall performance:** HRPm outperforms state-of-the-art DRL and classical portfolio strategies in both U.S. and Chinese markets.

Limitations

- Increased complexity from two-tier architecture requires higher training costs.
- Sensitivity to hyperparameters (e.g., entropy weight η) can shift balance between risk and return.
- Tested only in backtesting — lacks real-time trading validation.
- Execution dynamics may vary under different market conditions not covered in the study.

Contribution to My Project

3-3-4 Multi-type data fusion framework based on deep reinforcement learning for algorithmic trading [10]

Research Focus

The paper addresses the limitations of single-source DRL trading systems, which often fail to capture the complexity of financial markets. It proposes **MSF-DRL**, a multi-type data fusion framework that integrates raw stock data, technical indicators, and candlestick chart images to provide a richer, more comprehensive market representation.

Methodology

- **Data fusion pipeline:**
 - **Numeric branch:** Stock prices and technical indicators processed by an LSTM to extract temporal features and reduce noise.
 - **Visual branch:** Candlestick charts encoded using a CNN + BiLSTM to capture both spatial and sequential visual patterns.
 - **Fusion:** Combined features from both branches form the state representation.
- **RL decision module:** The fused state representation is input to a DRL trading agent (buy/sell/hold decisions).

Results

- Outperformed traditional baselines and single-modality DRL models.
- Demonstrated higher profitability and improved Sharpe ratios, reflecting stronger risk-adjusted returns.
- Effective across both Chinese stock datasets and selected S&P 500 stocks, suggesting cross-market robustness.

Limitations

- Computationally more complex due to dual-branch feature extraction and fusion.
- Performance may depend heavily on candlestick chart quality and preprocessing choices.
- Still limited to historical backtesting — no live trading validation.
- Fusion limited to three modalities; other information sources (e.g., sentiment/news data) were not considered.

Contribution to My Project

3-3-5 Market Sentiment-Aware Deep Reinforcement Learning Approach for Stock Portfolio Allocation [11]

Research Focus

This paper tackles the limitation of DRL-based portfolio allocation strategies that rely only on historical prices. It introduces a sentiment-aware DRL framework that integrates both market prices and investor sentiment—capturing psychology-driven market dynamics that strongly influence stock movements.

Methodology

- **Integrated inputs:** Combines historical stock data with sentiment signals extracted from news and social media.
- **Sentiment extraction:** Though the specific NLP pipeline is not detailed, sentiment scores are transformed into quantitative features.
- **DRL framework:** The sentiment-enhanced state representation is fed into a DRL agent (algorithm not explicitly specified, e.g., DQN/PPO possible).
- **Portfolio scope:** Focused on Dow Jones-listed companies.

Results

- Outperforms conventional DRL models relying only on price data.
- Achieves higher **Sharpe ratios** and **annualized returns**.
- Demonstrates improved robustness in portfolio allocation by incorporating sentiment-driven signals.

Limitations

- Sentiment extraction process is not fully transparent (pipeline details missing).
- Relies on quality and reliability of sentiment data, which may be noisy or biased.
- Study limited to Dow Jones companies; generalization to other markets remains untested.
- Evaluation based on backtesting, without real-time deployment validation.

Contribution to My Project

3-3-6 A multi-agent reinforcement learning framework for optimizing financial trading strategies based on TimesNet [12]

Research Focus

The study addresses the lack of robustness in single-agent DRL models for financial trading. It introduces **MADDQN**, a multi-agent reinforcement learning framework that combines diverse feature extractors—TimesNet and Multi-Scale CNN—to design trading strategies that balance return maximization and risk control.

Methodology

- **Dual-agent design:**
 - **Agent 1:** Uses **TimesNet**, transforming 1D time-series into 2D representations to capture periodic and multi-scale temporal patterns.
 - **Agent 2:** Uses a **Multi-Scale CNN** to capture temporal features at different resolutions.
- **Algorithmic backbone:** Both agents are trained with **Double DQN (DDQN)**, reducing Q-value overestimation bias.
- **Pre-training and fine-tuning:**
 - Pre-trained on a mixed dataset of major U.S. indices for generalization.
 - Fine-tuned on specific assets for scenario adaptation.

Results

- Achieved an average cumulative return of **23.08%** across five stock indices.
- Outperformed single-agent models and other baselines in terms of profitability and robustness.
- Demonstrated strong **generalization**: pre-training enabled transferability, while fine-tuning stabilized performance across assets.

- Multi-agent design produced more stable risk-return profiles than standalone DRL systems.

Limitations

- Computational overhead due to maintaining and coordinating multiple agents.
- Dependence on feature extractors (TimesNet, CNN) may limit adaptability in radically different market regimes.
- Tested only on historical backtests — no live trading experiments.
- Focused primarily on U.S. stock indices; cross-market scalability remains to be validated.

Contribution to My Project

3-4 Hybrid Approaches: Integrating LLMs with RL in Finance

3-4-1 FINCON: A Synthesized LLM Multi-Agent System with Conceptual Verbal Reinforcement for Enhanced Financial Decision Making [13]

Research Focus

FINCON addresses the complexity of sequential financial decision-making by combining LLMs with reinforcement learning in a **multi-agent system**. It introduces a manager–analyst hierarchy, conceptual verbal reinforcement, and dual-level risk control to produce more robust, interpretable, and risk-aware financial strategies.

Methodology

- **Multi-agent structure:**
 - **Market Analyst Agents:** Extract and summarize raw market data (e.g., stock prices, fundamentals).
 - **Sentiment & News Agents:** Process financial news, reports, and social media, converting unstructured text into sentiment insights.
 - **Strategy Agents:** Propose candidate trading or portfolio actions by integrating analyst and sentiment outputs.
 - **Risk Management Agents:** Monitor exposure, apply risk constraints, and manage portfolio-level safety.
 - **Manager (Decision-Maker) Agent:** Synthesizes all inputs and finalizes portfolio or trading decisions.
- **Risk control:**

- Within-episode → monitors CVaR daily to limit short-term risk.
- Over-episode → updates beliefs and strategies after each trading cycle.
- **Conceptual Verbal Reinforcement (CVRF):** Agents self-critique and share distilled “conceptual beliefs” in natural language to guide future decisions.
- **POMDP formulation:** Frames trading/portfolio tasks as POMDPs with latent factors (e.g., macro uncertainty).
- **Prompt optimization:** Uses text-based gradient descent to refine prompts across episodes for consistent reasoning.

Results

- Outperforms baseline DRL agents and other LLM-based approaches.
- Improves **cumulative return (CR)**, **Sharpe ratio (SR)**, and reduces **maximum drawdown (MDD)**.
- Manager–analyst hierarchy improves efficiency by reducing communication overhead.
- CVRF enhances interpretability and generalization in both single-stock and portfolio tasks.

Limitations

- Computationally demanding due to multiple specialized agents and multimodal inputs.
- Relies on accurate prompt engineering and high-quality datasets.
- Current validation limited to backtests; no live trading deployment.
- Scaling the agent network may introduce coordination complexity.

Contribution to My Project

Chapter 4

Research Methodology and System Design

Chapter 5

Implementation

Chapter 6

Experiments and Results

Chapter 7

Discussion and Analysis

Chapter 8

Conclusion and Future Work

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Appendix A

Appendix A: Title of Appendix

Appendix B

Appendix B: Title of Appendix